

Model Quantization

Model Quantization

1. Introduction
 2. Preparation
 - 2.1. Environment Preparation and Model Training
 - 2.2. Prepare Calibration Image Set and PT Model
 - 2.3. Convert ONNX Model
 - 2.4. Quantize Model
 - 3.1. Model Prediction
- Effect Preview

[!NOTE]

Note: This uses D-Robotics's OpenExplore package and BPU algorithm toolchain. For environment setup, please refer to the official D-Robotics GITHUB documentation. This article will not cover the environment setup process.

This example uses YOLO's official model yolo11n.pt, quantized to a .bin file

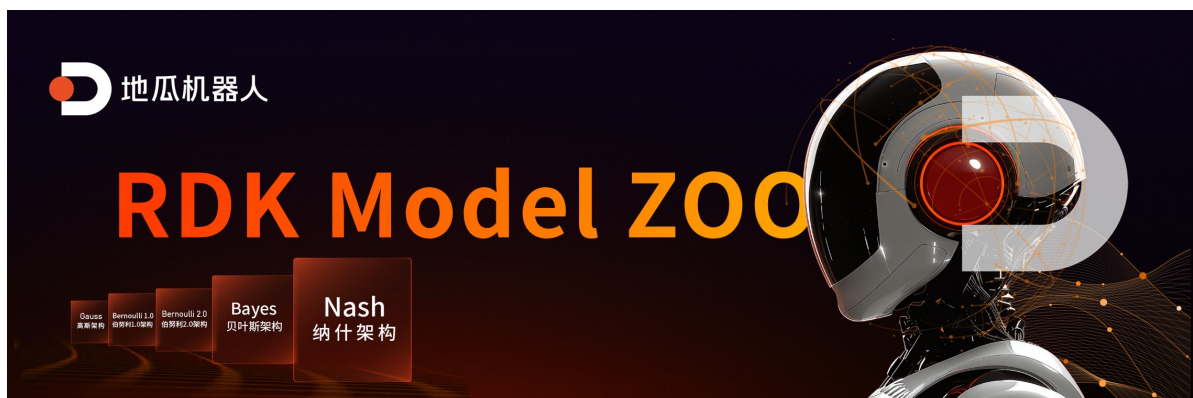
- The provided virtual machine `Rosmaster-M1-Ubuntu22.04` already has the OE package `horizon_x5_open_explorer_v1.2.8-py310_20240926` and `docker(ubuntu 20.04 cpu version)open_explorer_v1.2.8-py310_20240926` and environment set up.
- If you need to use GPU docker images, you need to set them up yourself according to the official documentation.

```
D-Robotics OpenExplore(RDK X5, Bayes-e BPU) Version: >= 1.2.8
Ultralytics YOLO Version: 8.3.193
Ubuntu 22.04
Python 3.10
```

1. Introduction

This article is written based on D-Robotics official GITHUB documentation. The prediction program is developed based on rdk_model_zoo. If you need to study and learn, you can search for it yourself.

- [Ultralytics YOLO: You Only Look Once](#)
- [Introduction to RDK Model Zoo](#)



- [Object Detection](#)



2. Preparation

2.1. Environment Preparation and Model Training

Note: This operation needs to be performed on your own x86 machine. It is recommended to use a machine with hardware acceleration for training, such as a GPU supporting CUDA, where `torch.cuda.is_available()` is True. Ubuntu 22.04, Python 3.10 environment is recommended.

Download the ultralytics/ultralytics repository and refer to the official ultralytics documentation to set up the environment.

For model training, please refer to the official ultralytics documentation. This document is maintained by ultralytics and is of very high quality. There are also many reference materials online. It is not difficult to obtain a model with pre-trained weights like the official one. Please note that no program needs to be modified during training, and the forward method does not need to be modified.

Ultralytics YOLO Official Documentation:

- Quick Start: <https://docs.ultralytics.com/quickstart/>
- Model Training: <https://docs.ultralytics.com/modes/train/>

Prepare the `rosmaster-M1-Ubuntu22.04` virtual machine from the learning materials!

2.2. Prepare Calibration Image Set and PT Model

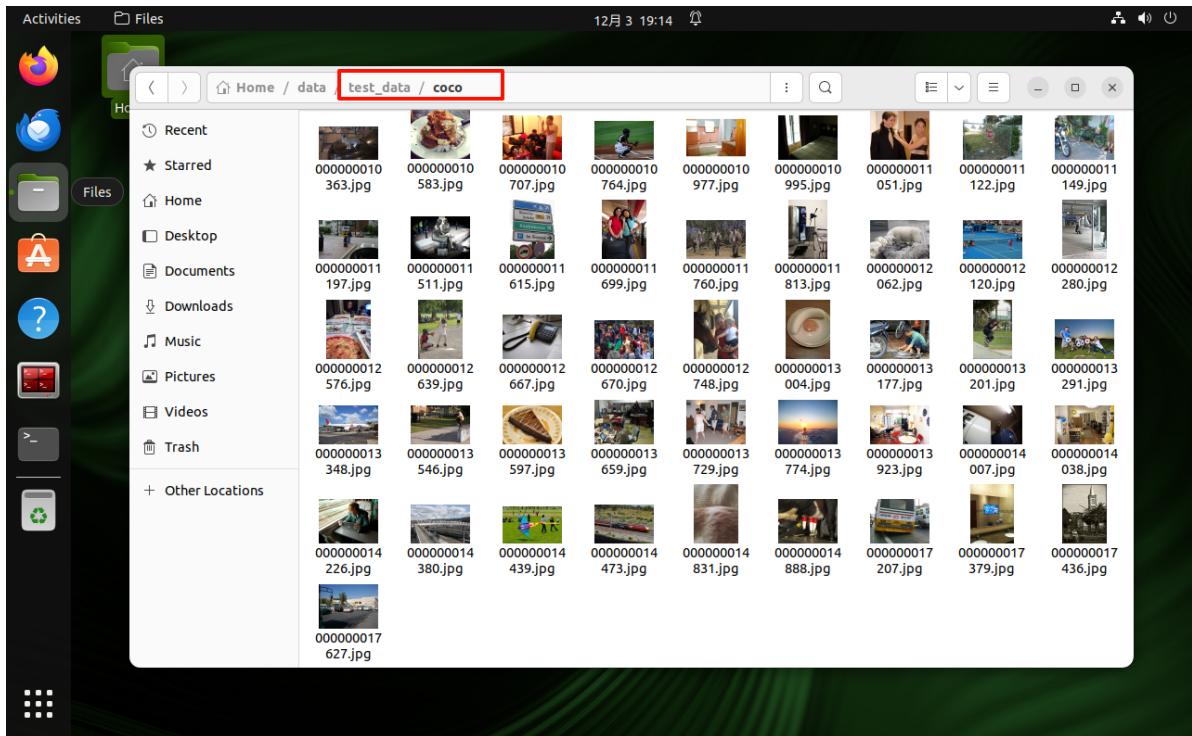
Download `va12017` from the official COCO dataset, and select about 100 images from the validation set - not too many, not too few.

[!TIP]

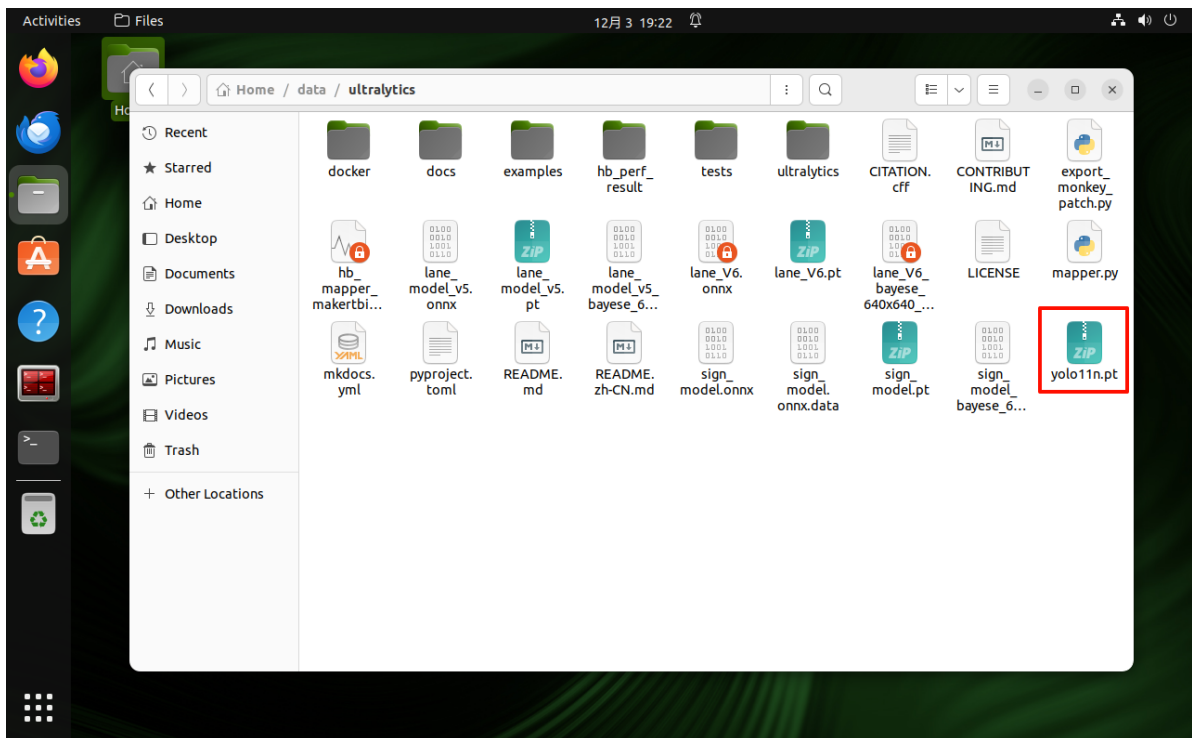
The entire preparation work is done within the virtual machine.

Use your own PT model, and use your own validation set images!

- Create a folder named `coco` yourself and put the images in it



- Put the `yolo11n.pt` model file in this path directory



2.3. Convert ONNX Model

- Open terminal and enter the algorithm toolchain docker container

```
~/rdk.sh
```

- Navigate to the previous path

```
cd ~/data/ultralytics/
```

```
python3 export_monkey_patch.py --pt yolo11n.pt
```

```
Activities Terminal 12月3 19:31
[System Information]
IP_Address_1: 192.168.11.194
IP_Address_2: 172.17.0.1
-----
ROS_DOMAIN_ID: 61 | ROS: humble
-----
yahboom@VM: ~$ ~/rdk.sh
root@b73cc00a7f94:~/data/ultralytics# cd ~/data/ultralytics/
root@b73cc00a7f94:~/data/ultralytics# python3 export_monkey_patch.py --pt yolo11n.pt
WARNING ⚠️ user config directory '/root/.config/Ultralytics' is not writeable, using '/tmp/Ultralytics'.
Set YOLO_CONFIG_DIR to override.
Creating new Ultralytics Settings v0.0.6 file ✓
View Ultralytics Settings with 'yolo settings' or at '/tmp/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
[Cauchy] Replaced Attention_forward in attn
[Cauchy] Replaced Detect_forward in z3
Ultralytics 8.3.214 Python-3.10.15 torch-1.13.0+cpu CPU (12th Gen Intel Core i5-12400)
⚡ ProfTip: Export to OpenVINO format for best performance on Intel hardware. Learn more at https://docs.
ultralytics.com/integrations/openvino/
YOLO11n summary (fused): 100 layers, 2,616,248 parameters, 0 gradients

PyTorch: starting from 'yolo11n.pt' with input shape (1, 3, 640, 640) BCHW and output shape(s) ((1, 80,
80, 80), (1, 80, 80, 64), (1, 40, 40, 80), (1, 40, 40, 64), (1, 20, 20, 80), (1, 20, 20, 64)) (5.4 MB)

ONNX: starting export with onnx 1.12.0 opset 11...
ONNX: export success ✓ 0.8s, saved as 'yolo11n.onnx' (10.0 MB)

Export complete (1.5s)
Results saved to /root/data/ultralytics
Predict:      yolo predict task=detect model=yolo11n.onnx imgsz=640
Validate:     yolo val task=detect model=yolo11n.onnx imgsz=640 data=/usr/src/ultralytics/ultralytics
/cfg/datasets/coco.yaml
Visualize:    https://netron.app
root@b73cc00a7f94:~/data/ultralytics#
```

The output shows successful export of the `yolo11n.onnx` model

- Algorithm toolchain model check

```
hb_mapper checker --model-type onnx --march bayes-e --model yolo11n.onnx
```

[!TIP]

According to the official introduction to this command, we input the command to check our model. The system will have a long output, and we can also see from the output that X5's BPU supports all YOLO operators, meaning all model calculations can be performed on X5's BPU. This check shows six output heads, so we can start the model conversion.

```
Activities Terminal 12月3 19:38
yahboom@VM: ~$ hb_mapper checker --model-type onnx --march bayes-e --model yolo11n.onnx
/model.23/cv3.2/cv3.2.0/cv3.2.0.0/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv3.2/cv3.2.0/cv3.2.0.0/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv3.2/cv3.2.0/cv3.2.0.1/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv3.2/cv3.2.0/cv3.2.0.1/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv3.2/cv3.2.1/cv3.2.1.0/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv3.2/cv3.2.1/cv3.2.1.0/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv3.2/cv3.2.1/cv3.2.1.1/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv3.2/cv3.2.1/cv3.2.1.1/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv3.2/cv3.2.2/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv2.2/cv2.2.0/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv2.2/cv2.2.0/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv2.2/cv2.2.1/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
/model.23/cv2.2/cv2.2.1/act/Mul BPU id(0) HzLut --
1.0 int8
/model.23/cv2.2/cv2.2.2/conv/Conv BPU id(0) HzSQuantizedConv --
1.0 int8
2025-12-03 19:36:02,008 INFO The quantized model output:
=====
Output
-----
2025-12-03 19:36:02,014 INFO End to Horizon NN Model Convert.
2025-12-03 19:36:02,015 INFO ONNX model output num : 6
2025-12-03 19:36:02,029 INFO End model checking....
root@b73cc00a7f94:~/data/ultralytics#
```

2.4. Quantize Model

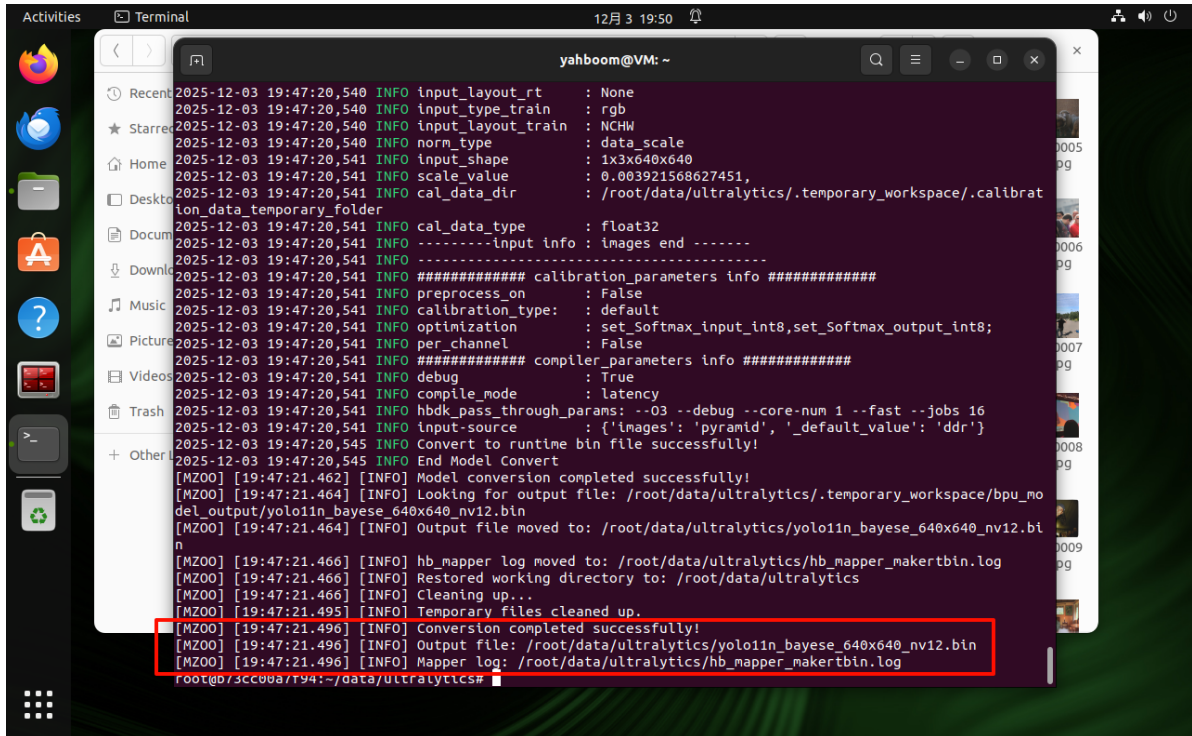
- Start model conversion

```
python3 mapper.py --onnx yolo11n.onnx --cal-images ~/data/test_data/coco
```

[!TIP]

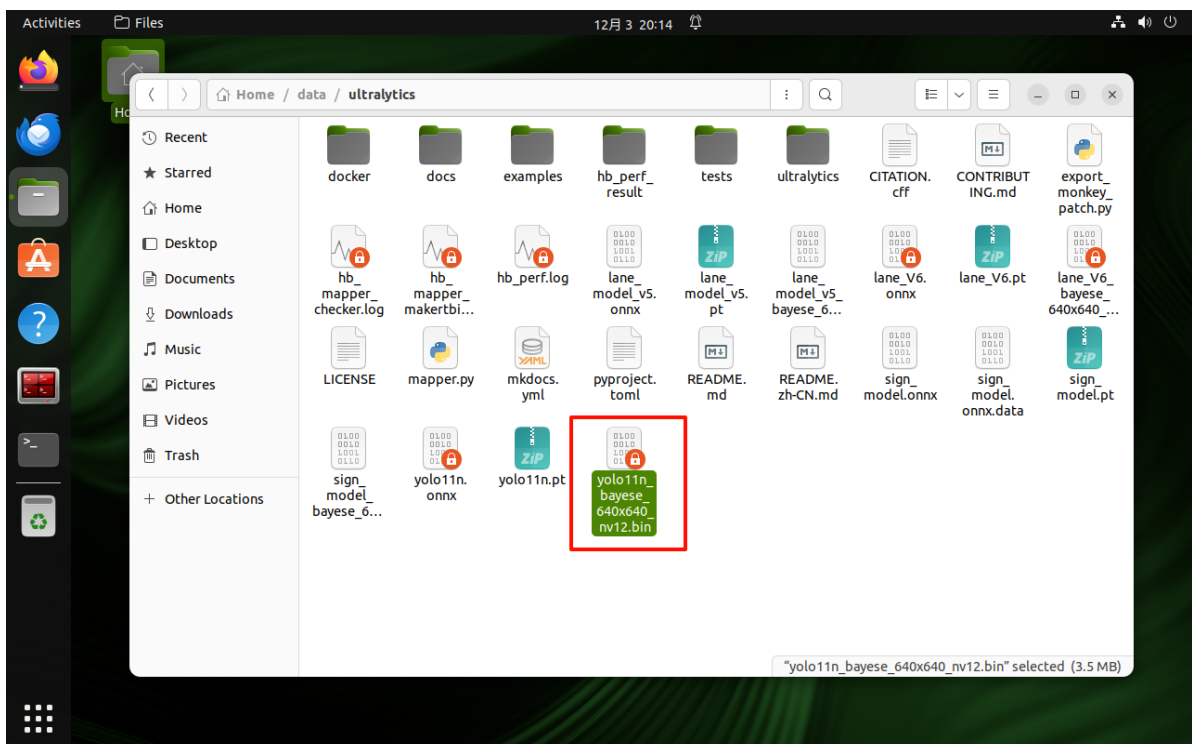
Fill in the path to your own dataset here

The conversion takes longer for CPU versions depending on model size, please be patient



```
2025-12-03 19:47:20,540 INFO input_layout_rt : None
2025-12-03 19:47:20,540 INFO input_type_train : rgb
2025-12-03 19:47:20,540 INFO input_layout_train : NCHW
2025-12-03 19:47:20,540 INFO norm_type : data_scale
2025-12-03 19:47:20,541 INFO input_shape : 1x3x640x640
2025-12-03 19:47:20,541 INFO scale_value : 0.003921568627451,
2025-12-03 19:47:20,541 INFO cal_data_dir : /root/data/ultralytics/.temporary_workspace/.calibrat
ion_data_temporary_folder
2025-12-03 19:47:20,541 INFO cal_data_type : float32
2025-12-03 19:47:20,541 INFO -----input info : images end -----
2025-12-03 19:47:20,541 INFO ##### calibration parameters info #####
2025-12-03 19:47:20,541 INFO preprocess_on : False
2025-12-03 19:47:20,541 INFO calibration_type : default
2025-12-03 19:47:20,541 INFO optimization : set_Softmax_input_int8,set_Softmax_output_int8;
2025-12-03 19:47:20,541 INFO per_channel : False
2025-12-03 19:47:20,541 INFO ##### compiler parameters info #####
2025-12-03 19:47:20,541 INFO debug : True
2025-12-03 19:47:20,541 INFO compile_mode : latency
2025-12-03 19:47:20,541 INFO hbdk_pass_through_params: --03 --debug --core-num 1 --fast --jobs 16
2025-12-03 19:47:20,541 INFO input-source : {'images': 'pyramid', 'default_value': 'ddr'}
2025-12-03 19:47:20,545 INFO Convert to runtime bin file successfully!
2025-12-03 19:47:20,545 INFO End Model Convert
[MZOO] [19:47:21.462] [INFO] Model conversion completed successfully!
[MZOO] [19:47:21.464] [INFO] Looking for output file: /root/data/ultralytics/.temporary_workspace/bpu_mo
del_output/yolo11n_bayese_640x640_nv12.bin
[MZOO] [19:47:21.464] [INFO] Output file moved to: /root/data/ultralytics/yolo11n_bayese_640x640_nv12.bi
n
[MZOO] [19:47:21.466] [INFO] hb_mapper log moved to: /root/data/ultralytics/hb_mapper_makerbin.log
[MZOO] [19:47:21.466] [INFO] Restored working directory to: /root/data/ultralytics
[MZOO] [19:47:21.466] [INFO] Cleaning up...
[MZOO] [19:47:21.495] [INFO] Temporary files cleaned up.
[MZOO] [19:47:21.496] [INFO] Conversion completed successfully!
[MZOO] [19:47:21.496] [INFO] Output file: /root/data/ultralytics/yolo11n_bayese_640x640_nv12.bin
[MZOO] [19:47:21.496] [INFO] Mapper log: /root/data/ultralytics/hb_mapper_makerbin.log
root@or-sc900a7194:~/data/ultralytics#
```

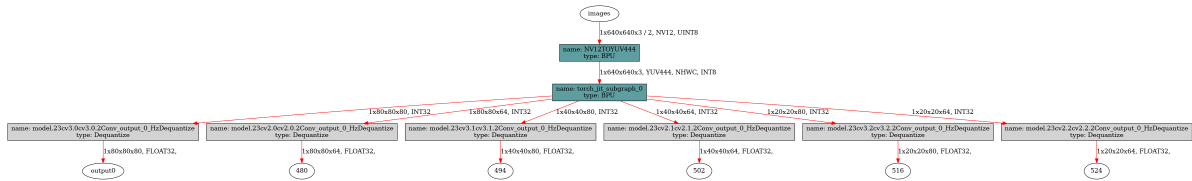
The output shows successful export of the `yolo11n_bayese_640x640_nv12.bin` model



- Algorithm toolchain model check


```
hb_perf yolol1n_bayese_640x640_nv12.bin
```

Then you will get a `hb_perf_result` folder with the verification results. We mainly focus on the model structure diagram inside. The figure below is for reference



3.1. Model Prediction

- (On the robot car host) Enter the ultralytics project

```
cd /home/sunrise/ultralytics/ultralytics
```

[!TIP]

Put the `yolol1n_bayese_640x640_nv12.bin` obtained from the virtual machine into the project directory on the robot car host

Before running the program, we need to modify the corresponding label names and model path in the program

```
# Need to modify
/home/sunrise/ultralytics/ultralytics/rdk_roscv.py
```

```
coco_names = [
    "person", "bicycle", "car", "motorcycle", "airplane", "bus", "train",
    "truck", "boat", "traffic light",
    "fire hydrant", "stop sign", "parking meter", "bench", "bird", "cat", "dog",
    "horse", "sheep", "cow",
    "elephant", "bear", "zebra", "giraffe", "backpack", "umbrella", "handbag",
    "tie", "suitcase", "frisbee",
    "skis", "snowboard", "sports ball", "kite", "baseball bat", "baseball
    glove", "skateboard", "surfboard", "tennis racket", "bottle",
    "wine glass", "cup", "fork", "knife", "spoon", "bowl", "banana", "apple",
    "sandwich", "orange",
    "broccoli", "carrot", "hot dog", "pizza", "donut", "cake", "chair", "couch",
    "potted plant", "bed",
    "dining table", "toilet", "tv", "laptop", "mouse", "remote", "keyboard",
    "cell phone", "microwave", "oven",
    "toaster", "sink", "refrigerator", "book", "clock", "vase", "scissors",
    "teddy bear", "hair drier", "toothbrush"
]
```

```

rdk_roscv.py U X
ultralytics > ultralytics > rdk_roscv.py
281 class RDKYOLONode(Node):
309     def image_callback(self, msg):
311         try:
386
387         except Exception as e:
388             self.get_logger().error(f"Error processing image: {e}")
389
390     coco_names = [
391         "person", "bicycle", "car", "motorcycle", "airplane", "bus", "train", "truck", "boat", "traffic light",
392         "fire hydrant", "stop sign", "parking meter", "bench", "bird", "cat", "dog", "horse", "sheep", "cow",
393         "elephant", "bear", "zebra", "giraffe", "backpack", "umbrella", "handbag", "tie", "suitcase", "frisbee",
394         "skis", "snowboard", "sports ball", "kite", "baseball bat", "baseball glove", "skateboard", "surfboard", "tennis racket", "bottle",
395         "wine glass", "cup", "fork", "knife", "spoon", "bowl", "banana", "apple", "sandwich", "orange",
396         "broccoli", "carrot", "hot dog", "pizza", "donut", "cake", "chair", "couch", "potted plant", "bed",
397         "dining table", "toilet", "tv", "laptop", "mouse", "remote", "keyboard", "cell phone", "microwave", "oven",
398         "toaster", "sink", "refrigerator", "book", "clock", "vase", "scissors", "teddy bear", "hair drier", "toothbrush"
399     ]
400 # coco_names = [
401 #     "greenlight", "honking", "parkA", "parkB", "redlight", "school",
402 #     "sidewalk", "sidewalk_ground", "speedlimit", "stop", "straight",
403 #     "turnright", "turnright_ground", "yellowlight"
404 # ]
405 # coco_names = [
406 #     "lane",
407 # ]

```

```

rdk_roscv.py U X
ultralytics > ultralytics > rdk_roscv.py
413 def draw_detection(img, bbox, score, class_id) -> None:
426     label = f"{coco_names[class_id]}: {score:.2f}"
427     (label_width, label_height), _ = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX, 0.5, 1)
428     label_x, label_y = x1, y1 - 10 if y1 - 10 > label_height else y1 + 10
429     cv2.rectangle(
430         img, (label_x, label_y - label_height), (label_x + label_width, label_y + label_height), color, cv2.FILLED
431     )
432     cv2.putText(img, label, (label_x, label_y), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 0), 1, cv2.LINE_AA)
433
434
435 def main():
436     parser = argparse.ArgumentParser()
437     parser.add_argument('--model-path', type=str,
438         default='/home/sunrise/ultralytics/ultralytics/yolo11n_bayese_640x640_nv12.bin',
439         #default='/home/sunrise/ultralytics/ultralytics/lane_V6_bayese_640x640_nv12.bin',
440         help="""Path to BPU Quantized *.bin Model.
441             RDK X3(Module): Bernoulli2.
442             RDK Ultra: Bayes.
443             RDK X5(Module): Bayes-e.
444             RDK S100: Nash-e.
445             RDK S100P: Nash-m.""")
446     parser.add_argument('--classes-num', type=int, default=14, help='Classes Num to Detect.')
447     parser.add_argument('--nms-thres', type=float, default=0.7, help='IoU threshold.')
448     parser.add_argument('--score-thres', type=float, default=0.25, help='confidence threshold.')
449     parser.add_argument('--reg', type=int, default=16, help='DFL reg layer.')
450     parser.add_argument('--strides', type=lambda s: list(map(int, s.split(','))),
451         default=[8, 16, 32],
452         help='--strides 8, 16, 32')
453     opt = parser.parse_args()
454     logger.info(opt)

```

Start the following programs in the terminal according to your camera model,

```

# USB camera
ros2 launch usb_cam camera.launch.py
# Nuwa camera
ros2 launch ascamera hp60c.launch.py

# Prediction program
python3 rdk_roscv.py

```

Effect Preview

[!TIP]

Taking Nuwa camera as an example,

You can see the inference speed is around 12ms, which is very impressive for the quantized model.

Applications: RDK YOLO Detection Software Updater sunrise@ubuntu: ~/ultralytics sunrise@ubuntu: ~ 125x19

sunrise@ubuntu: ~/ultralytics/ultralytics

ascamera_node-1 [INFO] [1764764304.402658365] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1148] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5480621] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1149] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5484879] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1150] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5485952] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1151] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5486565] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1152] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5487142] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1153] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5487833] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1154] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5488416] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1155] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5489055] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1156] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5489681] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1157] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5524039] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1158] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764304.5590726] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1159] [streamCallback] set gain ret 0, gain 1.0

ascamera_node-1 [INFO] [1764764343.3467488] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1160] [streamCallback] set gain ret 0, gain 1.0

[278] [stopStreaming] stop streaming

ascamera_node-1 [INFO] [1764764468.4987776] [ascamera_hp60c.camera_publisher]: 2025-12-03 20:18:24[INFO] [CameraHp60c.cpp] [1161] [streamCallback] set gain ret 0, gain 1.0

[259] [startStreaming] start streaming

[BPU_PLAT]BPU Platform Version(1.3.6)!

[HBRT] set log level as 0. version = 3.15.55

[DNN] Runtime version = 1.24.5 (3.15.55 HBRT)

[A][DNN][packed_model.cpp:247][Model](2025-12-03 20:18:24)

[RDK_YOLO] [20:21:06.784] [INFO] -> input tensor

[RDK_YOLO] [20:21:06.784] [INFO] input[0], n

[RDK_YOLO] [20:21:06.785] [INFO] -> output tensor

[RDK_YOLO] [20:21:06.785] [INFO] output[0], n

[RDK_YOLO] [20:21:06.785] [INFO] output[1], n

[RDK_YOLO] [20:21:06.785] [INFO] output[2], n

[RDK_YOLO] [20:21:06.785] [INFO] output[3], n

[RDK_YOLO] [20:21:06.785] [INFO] output[4], n

[RDK_YOLO] [20:21:06.786] [INFO] output[5], n

[RDK_YOLO] [20:21:06.786] [WARNING] Provided classes_num=14 but model output indicates 80

[RDK_YOLO] [20:21:06.786] [INFO] Using classes_num = 80

[RDK_YOLO] [20:21:06.789] [INFO] Model initialized, input size=(640, 640), strides=[8, 16, 32]

[INFO] [1764764467.202351575] [rdk_yolo_node]: Using camera topic: /ascamera_hp60c/camera_publisher/rgb0/image

[INFO] [1764764467.516659146] [rdk_yolo_node]: Using camera topic: /ascamera_hp60c/camera_publisher/rgb0/image

RDK YOLO Detection

FPS: 11.42
Infer: 12.7ms
Pre: 5.6ms Post: 0.5ms
Detections: 5
Total: 28.8ms

person: 0.32
tv: 0.47
tv: 0.67
chair: 0.76

(x=261, y=62) - R:28 G:40 B:38